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ASYMPTOTIC BOUNDS FOR THE NUMBER OF CONVEX n-OMINOES*

David A. KLARNER and Ronald L. RIVEST

Computer Science Department, Stanford University, Stanford, Calif. 94305, USA

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Abstract. Unit squares having their vertices at integer points in the Cartesian plane are called *cells*. A point set equal to a union of n distinct cells which is connected and has no finite cut set is called an *n-omino*. Two n-ominoes are considered the same if one is mapped onto the other by some translation of the plane. An n-omino is *convex* if all cells in a row or column form a connected strip. Letting $c(n)$ denote the number of different convex n-ominoes, we show that the sequence $((c(n))^{1/n}: n = 1, 2, \ldots)$ tends to a limit γ, and $\gamma = 2.309138\ldots$.

1. Introduction

Unit squares having their vertices at integer points in the Cartesian plane are called *cells*. A point set equal to a union of n distinct cells which is connected and has no finite cut set is called an *n-omino*. Two n-ominoes are considered the same if one is mapped onto the other by some translation of the plane. (Such n-ominoes were called *fixed* animals with n cells by Read [8].) For example, there are six different 3-ominoes as shown in Fig. 1.

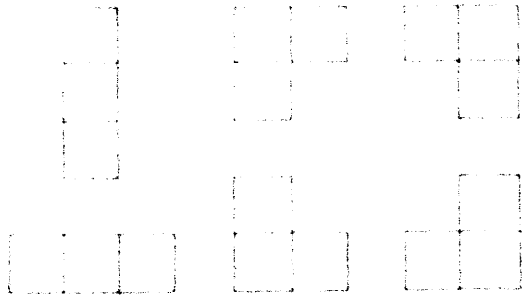

Fig. 1. The 3-ominoes.

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Let $t(n)$ denote the number of distinct n-ominoes. It is known [3] that the sequence $((t(n))^{1/n} : n = 1, 2, \ldots)$ tends to a limit θ. The investigation of θ began with Eden's work [1]; he managed to prove that $3.14 < \theta \leq 6.75$. There has been considerable effort expended to improve these bounds. Currently, the best lower bound (given in [3]) is $3.72 < \theta$, while the best upper bound (given in [5]) is $\theta < 4.65$.

An n-omino is *row-convex* when each row of the n-omino is a connected strip of cells. *Column-convex* n-ominoes are defined analogously. All six of the 3-ominoes (shown in Fig. 1) are both row-convex and column-convex; in general such n-ominoes are said to be *row-column-convex*, or just *convex* for short. It was shown in [2] (and in [3] by a second method) that

$$(1) \qquad \frac{x(1-x)^3}{1-4x+7x^2-5x^3} = \sum_{n=1}^{\infty} b(n)\, x^n ,$$

where $b(n)$ denotes the number of distinct row-convex n-ominoes. (This result was also obtained by Pólya [6].) Thus it follows that the sequence $((b(n))^{1/n} : n = 1, 2, \ldots)$ tends to a limit β which is equal to the largest real root of $y^3 - 4y^2 + 7y - 5 = 0$; that is, $\beta = 3.20\ldots$.

Recently, Donald Knuth wrote us and asked us if the number $c(n)$ of convex n-ominoes had been investigated. This paper is entirely motivated by Knuth's question. We shall be concerned with the problem of effectively calculating the limit γ of the sequence $((c(n))^{1/n} : n = 1, 2, \ldots)$. One of the first things we prove is that this limit exists. Later on we show how to calculate upper and lower bounds for γ and give the best results obtained by these methods.

2. Existence of $\lim_{n\to\infty} (c(n))^{1/n}$

Following Ceasar's admonition, we divide, then conquer. A convex n-omino may be split into three parts by making two cuts between certain rows so that the upper and lower parts are roughly trapezoids and the middle part is roughly a parallelogram. A typical sectioning of this sort is shown in Fig. 2. More precisely, the trisection of a convex n-omino A is accomplished by cutting along the lowest level of A where the left boundary of A goes to the right and by cutting along the lowest level of A where the right boundary of A goes to the left.

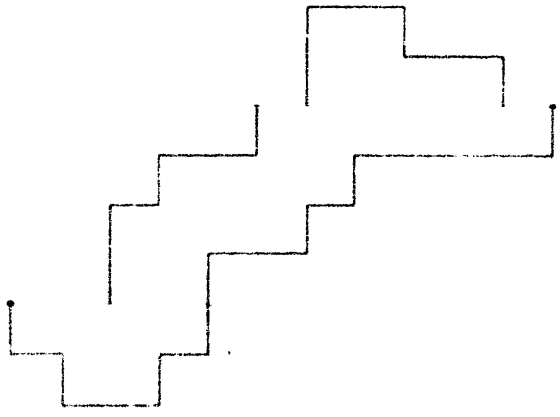

Fig. 2. Trisection of a convex 28-omino.

A convex n-omino whose left boundary climbs to the right and whose right boundary climbs to the left corresponds to a partition of n called a *stack* by Wright [9]. We let $s(n)$ denote the number of distinct n-ominoes corresponding to stacks; for example, there are four 3-ominoes shown in Fig. 1 which correspond to stacks, so $s(3) = 4$. A convex n-omino whose left and right boundaries both climb to the right is called a *parallelogram*, and $p(n)$ will denote the number of distinct n-ominoes which are parallelograms. Clearly, $p(n) \leq c(n)$ for all n; also, $s(n) \leq p(n)$ for all n (the diagram in Fig. 3 suggests a proof of this fact). Finally, an obvious construction establishes that $p(m)\,p(n) \leq p(m+n)$ for all m, n. Now we use the fact that if $\{u_n\}$ is a sequence of natural numbers such that $((u_n)^{1/n} : n = 1, 2, \ldots)$ is bounded and $u_m\, u_n \leq u_{m+n}$ for all m, n, then $\lim_{n\to\infty} (u_n)^{1/n}$ exists. (For similar results, see Pólya and Szegö

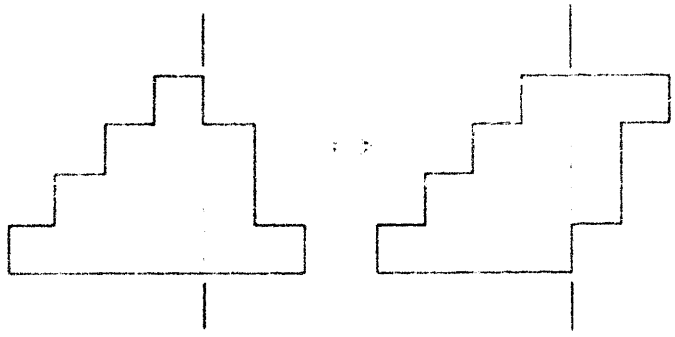

Fig. 3. An injection showing $s(n) \leq p(n)$.

[7, p. 171].) We have $p(n) \le b(n) < (3.20)^n$ for all large n, and $p(m)p(n) \le p(m+n)$, so

$$(2) \qquad \lim_{n\to\infty} (p(n))^{1/n} = \gamma$$

exists. Using the fact that every convex n-omino splits into two stacks and one parallelogram, we can reconstruct these n-ominoes by pasting together two stacks and one parallelogram in various ways. Again, using an obvious construction, and using the fact that $p(i)p(j)p(k) \le p(i+j+k)$ for all i, j, k, it is easy to show that

$$(3) \qquad c(n) \le 2n^2 \sum_{(i,j,k)} s(i)p(j)s(k) \le 2n^2 \sum_{(i,j,k)} p(i)p(j)p(k)$$

$$\le 2n^2 \binom{n+2}{2} p(n) \le (n+2)^4 p(n),$$

where the index of summation in the sums extends over all compositions (i, j, k) of n into non-negative parts. There are $\binom{n+2}{2}$ such compositions.

Using (2) and (3) together with the fact that $p(n) \le c(n)$ for all n, we have

$$(4) \qquad \gamma = \lim_{n\to\infty} (p(n))^{1/n} \le \liminf_{n\to\infty} (c(n))^{1/n}$$

$$\le \limsup_{n\to\infty} (c(n))^{1/n} \le \lim_{n\to\infty} ((n+2)^4 p(n))^{1/n} = \gamma .$$

Hence $\lim_{n\to\infty} (c(n))^{1/n}$ exists, and

$$(5) \qquad \lim_{n\to\infty} (c(n))^{1/n} = \lim_{n\to\infty} (p(n))^{1/n} = \gamma .$$

3. An integral equation

We shall use a theory developed in [4] concerning a double sequence $(b(n, a): n, a = 1, 2, \ldots)$ defined in terms of given sequences $(f(m, n): m, n = 1, 2, \ldots)$ and $(g(n): n = 1, 2, \ldots)$ as follows:

$$(6) \qquad b(n, a) = \sum f(a_1, a_2) f(a_2, a_3) \ldots f(a_{k-1}, a_k) g(a_k) .$$

where the index of summation extends over all k-tuples $(a_1, \ldots, a_k)$ of natural numbers for $k = 1, \ldots, n$ with $a_1 = a$ and $a_1 + \ldots + a_k = n$. It was shown that if

$$(7) \qquad G(x) = \sum_{n=1}^{\infty} g(n) x^n$$

and

$$(8) \qquad F(x, y) = \sum_{m,n=1}^{\infty} f(m, n) x^m y^n$$

converge for $|x|$ and $|y|$ sufficiently small, then

$$(9) \qquad B(x, y) = \sum_{n=1}^{\infty} \sum_{a=1}^{n} b(n, a) y^a x^n$$

converges for $|x|$ and $|y|$ sufficiently small, and

$$(10) \qquad B(x, y) = G(xy) + \frac{1}{2\pi i} \int_C F(xy, 1/s) B(x, s) \frac{ds}{s} ,$$

where C is a contour in the s-plane which includes $s = 0$ and the singularities of $F(xy, 1/s)$ but excludes the singularities of $B(x, s)$. The theory of (10) runs parallel to that of the Fredholm integral equation. In particular, if $F(x, y)$ has the special form

$$(11) \qquad F(x, y) = R_1(x) S_1(y) + \ldots + R_t(x) S_t(y) ,$$

we say F is *separable*, and it turns out that (10) can be converted into a system of t equations linear in t unknown functions. The system can be solved and the solution yields a formula for $B(x, y)$. We shall give an example of this later on.

If F is not separable, we can still get information about B by approximating F with something that is separable. Suppose

$$(12) \qquad K(x, y) = \sum k(m, n) x^m y^n$$

and $k(m, n) \leq f(m, n)$ for all m, n, then we say K is *a lower bound* on F, an *upper bound* on F is defined analogously. If K is separable, we may

substitute K for F in (10) and calculate a lower bound for B. Upper bounds for B may be obtained in a similar fashion. We shall adopt this strategy too, so an example is forthcoming.

The relevance of the foregoing discussion to the enumeration of n-celled parallelograms is as follows: the number of $(m+n)$-celled parallelograms having m cells in one row and n cells in a second row is

$$f(m, n) = \min\{m, n\}\,. \tag{13}$$

It is fairly easy to show that the number of n-celled parallelograms with exactly k rows of cells having exactly a_i cells in the ith row for $i = 1, \ldots, k$ is

$$f(a_1, a_2) f(a_2, a_3) \ldots f(a_{k-1}, a_k)\,. \tag{14}$$

Thus, if we take f as defined in (13) and put $g(j) = 1$ for all j, we can sum (6) over $a = 1, \ldots, n$ and obtain $p(n)$. In this case, we have

$$F(x, y) = xy/(1-x)(1-y)(1-xy)\,, \tag{15}$$

$$G(x) = x/(1-x)\,. \tag{16}$$

Substituting these functions in (10) gives

$$\begin{aligned} B(x, y) &= \frac{xy}{1-xy} + \frac{1}{2\pi i}\int_C \frac{xy\,B(x, s)\,ds}{(1-xy)(s-1)(s-xy)} \\ &= \frac{xy}{1-xy} + \frac{xy}{(1-xy)^2}\,B(x, 1) - \frac{xy}{(1-xy)^2}\,B(x, xy)\,. \end{aligned} \tag{17}$$

We can iterate (17) to eliminate $B(x, xy)$, $B(x, x^2y)$, ... successively to find

$$B(x, y) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}\, x^{k(k+1)/2}\, y^k (1 - x^k y + B(x, 1))}{(1-xy)^2 (1-x^2y)^2 \ldots (1-x^k y)^2}\,. \tag{18}$$

Setting $y = 1$ in (18), we solve for $B(x, 1)$, the generating function of $(p(n): n = 1, 2, \ldots)$, which turns out to be

$$(19)\quad B(x,1)=\frac{\dfrac{x}{1-x}-\dfrac{x^3}{(1-x)^2(1-x^2)}+\dfrac{x^6}{(1-x)^2(1-x^2)^2(1-x^3)}-\cdots}{1-\dfrac{x}{(1-x)^2}+\dfrac{x^3}{(1-x)^2(1-x^2)^2}-\dfrac{x^6}{(1-x)^2(1-x^2)^2(1-x^3)^2}+\cdots}$$

$$=\sum_{n=1}^{\infty}p(n)x^n\ .$$

We have been unable to make use of (19) in estimating $p(n)$. Instead, we use upper and lower bounds for F as defined in (15), and then use (10) to calculate upper and lower bounds for B.

4. Lower bounds

Let

$$(20)\qquad F_k(x,y)=\sum_{m,n=1}^{k}f(m,n)x^m y^n\ ,$$

where $f(m,n)=\min\{m,n\}$ just as in (13), and let $B_k(x,y)$ denote the solution of (10) having F_k substituted for F. Since F_k is a lower bound for F, it follows that B_k is a lower bound for B. It was shown in [4] that when the kernel of (10) is approximated by a polynomial as in this case, then $B_k(x,1)$ is a rational function, say $B_k=P_k/Q_k$ with P_k and Q_k polynomials, and the denominator of B_k may be expressed as a determinant. In the present situation this turns out to be

$$(21)\qquad Q_k(x)=\begin{vmatrix} 1-x & 1 & 1 & \dots & 1\\ 1 & 2-x^2 & 2 & \dots & 2\\ 1 & 2 & 3-x^3 & \dots & 3\\ \vdots & \vdots & \vdots & \vdots & \vdots\\ 1 & 2 & 3 & \dots & k-x^k \end{vmatrix}\ .$$

If we put $Q_0(x)=1$ and $Q_1(x)=1-x$, we can use (21) to verify that

$$(22)\qquad Q_k(x)=(1-x^{k-1}-x^k)Q_{k-1}(x)-x^{2k-2}Q_{k-2}(x)$$

for $k = 2, 3, \ldots$. For example,

$$Q_2(x) = 1 - 2x - x^2 + x^3 ,$$
$$Q_3(x) = 1 - 2x - 2x^2 + 2x^3 + 2x^4 + x^5 - x^6 ,$$
$$Q_4(x) = 1 - 2x - 2x^2 + x^3 + 3x^4 + 5x^5 - 2x^6 - 2x^7 - 2x^8 - x^9 + x^{10} .$$

Letting γ_k denote the largest real root of $Q_k(1/x) = 0$, we have $\gamma_1 \leq \gamma_2 \leq \ldots \leq \gamma$, where γ is defined in (2). We have used a computer to calculate lower bounds for $\gamma_1, \gamma_2, \ldots, \gamma_{10}$ given in Table 1. Our results indicate that the sequence $\{\gamma_i\}$ converges very quickly to the value 2.30913859..., our best lower bound for γ.

Table 1

k	γ_k	β_k
1	1.00000000	2.41421356
2	2.24697960	2.33578290
3	2.30855218	2.31475605
4	2.30913772	2.31023504
5	2.30913859	2.30934711
6	2.30913859	2.30917790
7	2.30913859	2.30914598
8	2.30913859	2.30913998
9	2.30913859	2.30913885
10	2.30913859	2.30913864

5. Upper bounds

For $k = 1, 2, \ldots$, we define upper bounds $f^k(m, n)$ for $f(m,n) = \min\{m, n\}$ as follows:

$$(25) \qquad f^k(m, n) = \begin{cases} m & \text{if } k < n < m, \\ f(m, n) & \text{otherwise.} \end{cases}$$

Hence

$$(24) \qquad F^k(m, n) = \sum_{m,n=1}^{\infty} f^k(m, n)\, x^m y^n$$

$$= \frac{xy}{(1-x)^2(1-y)} - \frac{x^2 y}{(1-x)^2} - \cdots - \frac{x^{k+1} y^k}{(1-x)^2}$$

is an upper bound for F; furthermore, note that F^k is separable.

Let B^k denote the solution of (10) with F^k substituted for F. Then

$$(25)\qquad B^k(x,y) = \frac{xy}{1-xy} + \frac{xy\,B^k(x,1)}{(1-xy)^2} - \frac{xy}{(1-xy)^2}\sum_{r=1}^{k} x^r y^r B_r^k(x),$$

where

$$B_r^k(x) = \frac{1}{r!}\frac{\partial^r}{\partial s^r} B^k(x,s)\Big|_{s=0}.$$

Now we use (25) to get a system of equations involving $B_1^k, \ldots, B_k^k$. Take the rth partial derivative with respect to y at $y = 0$ and divide by $r!$ in (25) to get

$$(26)\qquad B_r^k(x) = x^r + r\,x^r B^k(x,1) - \sum_{j=1}^{r-1} (r-j)\,x^r B_j^k(x),$$

from which it follows that

$$(27)\qquad B_{r+1}^k(x) = (2x - x^{r+1})B_r^k(x) - x^2 B_{r-1}^k(x).$$

Setting $B_r^k(x) = P_r(x) + Q_r(x)B^k(x,1)$ for $r = 1, \ldots, k$, it follows that P_r and Q_r also satisfy the difference equation (27). Also we can substitute $P_r + Q_r B^k$. For B_r in (25) with $y = 1$ and solve for $B^k(x,1)$ in terms of $P_1, Q_1, \ldots, P_k, Q_k$ to obtain

$$(28)\qquad B^k(x,1) = \frac{x - x^2 - \sum_{j=1}^{k} x^{j+1} P_j(x)}{1 - 3x + x^2 + \sum_{j=1}^{k} x^{j+1} Q_j(x)}.$$

Thus B^k is a rational function whose numerator N_k and denominator D_k we know how to compute because they are defined in terms of $P_1, \ldots, P_k$ and $Q_1, \ldots, Q_k$ which we know how to compute. Let β_k denote the largest real root of $D_k(1/x)$, then we know

$$(29)\qquad \lim_{n\to\infty}\left(\sum_{a=1}^{n} b^k(n,a)\right)^{1/n} = \beta_k \le \gamma,$$

and $\beta_1 \ge \beta_2 \ge \ldots \ge \gamma$. Thus we can calculate upper bounds for $\beta_1, \beta_2, \ldots$ to obtain successively better upper bounds for γ.

Using the definitions

$$D_k = 1 - 3x + x^2 + x^2 Q_1 + \ldots + x^{k+1} Q_k , \tag{30}$$

$$Q_{r+1} = (2x - x^{r+1}) Q_r - x^2 Q_{r-1} \qquad (r > 1) , \tag{31}$$

and $Q_1 = x$, $Q_2 = 2x^2 - x^3$, the polynomials $D_1, D_2, \ldots$ are calculated with relative ease. For example, we found

$$D_1 = 1 - 3x + x^2 + x^3 ,$$
$$D_2 = 1 - 3x + x^2 + x^3 + 2x^5 - x^6 ,$$
$$D_3 = 1 - 3x + x^2 + x^3 + 2x^5 - x^6 + 3x^7 - 2x^8 - 2x^9 + x^{10} .$$

Using a computer, the polynomials $D_1, \ldots, D_{10}$ were calculated via (30) and upper bounds for β_k, the largest real root of $D_k(1/x) = 0$, were computed for $1 \le k \le 10$ using the Newton–Raphson method. These upper bounds for β_k are given in Table 1.

Combining our upper and lower bounds, we can conlude that

$$\gamma = \lim_{n \to \infty} (c(n))^{1/n} = 2.309138\ldots . \tag{32}$$

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